Research on Precision Education Management Practices in Vocational Colleges Based on Student Profiling

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Abstract

The application of student profiling technology has injected new vitality into vocational education management, with the prospect of realizing precise and personalized educational services. This paper first reviewed the theoretical foundation of student profiling technology, including its concepts, characteristics, and theoretical models. On this basis, it discussed the data collection and analysis methods of student profiles in vocational colleges, laying the foundation for subsequent practical applications. Furthermore, the paper focused on analyzing the precision education management practices based on student profiling, including personalized teaching models, academic early warning prediction models, and career planning support models. The application of these models has effectively improved the academic performance, warning accuracy, satisfaction, and graduation rates of vocational college students. Finally, the paper comprehensively evaluated the application effectiveness of student profiling technology in vocational colleges and looked forward to future development trends, proposing relevant suggestions.

CCS Concepts

• **Applied computing** → Education; Learning management systems.

Keywords

Profiling Technology, Vocational Education, Academic Performance, Effectiveness Evaluation, Academic Early Warning

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1 Introduction

1.1 Characteristics of Vocational College Students and Their Education Management Needs

Vocational colleges play a crucial role in China's higher education landscape, primarily tasked with nurturing high-quality technical and skilled professionals. However, the student population in these institutions presents unique characteristics that pose distinct challenges to educational management, setting them apart from their counterparts in general universities.[1] A primary concern is the relatively weak academic foundation of vocational college students, often attributed to their background in secondary vocational or technical schools, which can impede their progress in more advanced professional courses. This academic gap is further compounded by a noticeable lack of enthusiasm and interest among some students towards their chosen majors, leading to diminished learning initiative and suboptimal academic outcomes. Additionally, vocational college students often grapple with balancing theoretical knowledge and practical skills, face more immediate employment pressures, and may struggle with self-perception and career orientation. The societal stigma sometimes associated with vocational education can also impact students' confidence and motivation. To address these multifaceted challenges, vocational colleges must implement targeted strategies, including reinforcing foundational education, stimulating learning interest through practical experiences, offering personalized education and career guidance, strengthening industry-education integration, providing psychological support, innovating teaching methodologies, and assisting with comprehensive career planning.[2]

1.2 The Application Status and Significance of Student Profiling Technology in Vocational Education

Student profiling technology represents a cutting-edge approach to educational management, leveraging the power of big data and artificial intelligence to revolutionize how institutions understand and support their students.[3] This innovative methodology constructs comprehensive models of individual student characteristics by aggregating and analyzing a wide array of data points, including academic performance, psychological profiles, social interactions, and extracurricular engagements.[4] The resulting profiles offer a nuanced, multidimensional view of each student, enabling educators to tailor their approaches to meet diverse learning needs more

effectively. In the realm of vocational education, where student populations often present unique challenges, the adoption of this technology is gaining significant traction, offering institutions a powerful tool to enhance their educational strategies.[5] Pioneering vocational colleges implementing student profiling systems have reported promising outcomes, particularly in the realm of personalized learning. By harnessing the insights gleaned from these profiles, educators can craft bespoke teaching plans that address the specific strengths, weaknesses, and learning styles of different student segments. This data-driven approach not only boosts the relevance and efficacy of instruction but also fosters a more engaging and supportive learning environment.Furthermore, student profiling technology facilitates early intervention strategies, allowing institutions to identify and assist at-risk students before academic issues escalate. It also aids in career guidance by aligning students' aptitudes and interests with potential career paths, thereby enhancing overall student satisfaction and employability outcomes.

2 Theoretical Foundation and Practical Application of Student Profiling Technology

2.1 Concepts, Characteristics, and Theoretical Models of Student Profiling Technology

Student profiling technology represents a paradigm shift in educational management, harnessing the power of big data and artificial intelligence to create a holistic understanding of each student's unique characteristics and needs. This innovative approach constructs detailed models of individual students by systematically collecting and analyzing a wide array of information, encompassing academic performance, psychological traits, social interactions, and extracurricular engagements. At its core, this technology aims to synthesize and interpret multi-dimensional student data, resulting in comprehensive profiles that offer unprecedented insights into learners' strengths, challenges, and potential. The profiles generated are remarkably multifaceted, covering various aspects of a student's life and learning journey, including but not limited to academic achievements, learning styles, emotional intelligence, social skills, and personal interests. Crucially, these profiles are not static snapshots but dynamic, evolving representations that are continuously updated to reflect the student's growth, changing circumstances, and new information. This real-time adaptability ensures that the profiles remain relevant and accurate throughout a student's educational journey. The true power of student profiling technology lies in its ability to inform and guide targeted educational interventions and personalized learning strategies.

2.2 Data Collection and Analysis Methods for Student Profiling in Vocational Colleges

The construction of comprehensive student profiles in vocational colleges necessitates a multifaceted approach to data collection and analysis, encompassing a wide spectrum of information sources. This process involves gathering diverse data sets, including academic performance metrics, psychological assessments, socioeconomic background information, and career development indicators. Once collected, this rich tapestry of data undergoes sophisticated analysis using advanced data mining techniques and machine learning algorithms, enabling educators to uncover hidden patterns, correlations, and insights that might otherwise remain obscure. For instance, cluster analysis proves invaluable in identifying distinct student subgroups with shared characteristics, facilitating more targeted interventions and support mechanisms. Association rule analysis delves into the intricate relationships between various student attributes, revealing how different factors interplay to influence academic and personal outcomes. Predictive modeling takes this a step further, leveraging historical data to forecast future academic performance and career trajectories, allowing for proactive measures to support student success. The implementation of student profiling technology represents a paradigm shift in educational management for vocational colleges, offering a data-driven approach to understanding and nurturing student potential. By systematically collecting, analyzing, and interpreting comprehensive student data, institutions can gain unprecedented insights into individual and collective student needs, preferences, and challenges.This enhanced understanding empowers schools to design and implement highly targeted educational strategies, personalized learning pathways, and tailored support systems.

3 Precision Education Management Practices Based on Student Profiling

3.1 Personalized Teaching Model

Based on student profiling technology, schools can establish personalized teaching models to provide differentiated teaching services for students with different characteristics. As shown in Figure 1.

First, the school can use methods such as classification and clustering to divide students into different groups based on information such as their learning foundation, cognitive characteristics, and learning preferences. For example, students can be divided into groups such as excellent learners, average learners, and struggling learners.

Secondly, the school can design personalized teaching plans for different student groups. For excellent learners, heuristic teaching can be used to stimulate their exploration and innovation spirit; for average learners, hierarchical teaching can be used to provide guidance on their weak points; for struggling learners, remedial teaching can be used to help them make up for the lack of basic knowledge and skills.

3.2 Academic Early Warning Prediction Model

Student profiling technology can also be used to establish an academic early warning prediction model to help schools timely identify and intervene with students at academic risk. First, the school can collect data on students' academic performance, classroom performance, and assignment completion, combined with information on their psychological characteristics and family background, to establish an academic early warning prediction model. This model can use machine learning algorithms such as logistic regression and decision trees to predict the probability of students' academic risk. Secondly, the school can take targeted intervention measures for students of different risk levels based on the analysis results of the early warning model. For high-risk students, the school can arrange specialized tutors to provide one-on-one assistance; for

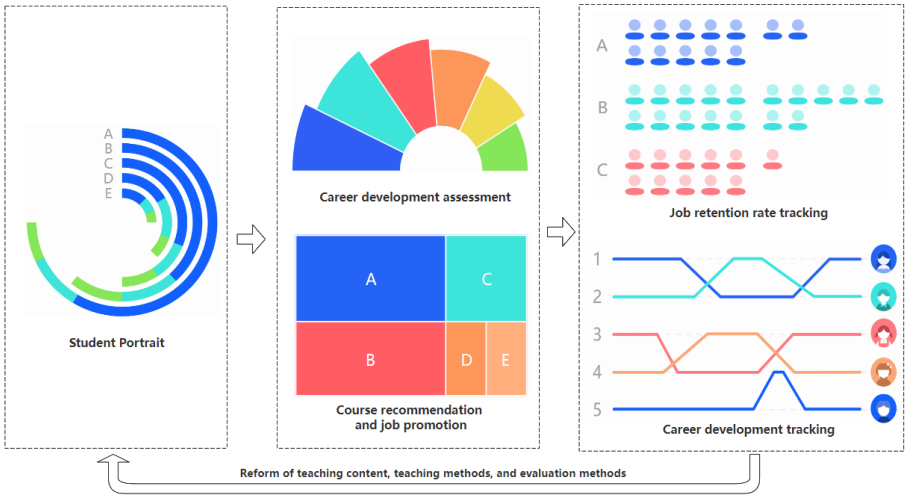


Figure 1: Precision Education Management Practice Model Based on Student Profiling.

medium-risk students, the school can adopt supplementary training and psychological counseling; for low-risk students, the school can provide appropriate incentives to promote their continuous progress.

3.3 Career Planning Support Model

Student profiling technology can be effectively applied to the construction of comprehensive career planning support models, significantly enhancing students' ability to clarify their career development directions and substantially improve their employment quality. This innovative approach leverages data-driven insights to create a more personalized and effective career guidance system.

Through sophisticated student profiling techniques, educational institutions can gain a deep and nuanced understanding of each student's unique combination of interests, personality traits, and professional strengths. This detailed profiling goes beyond traditional assessment methods, incorporating a wide range of data points collected throughout the student's academic journey. By analyzing this rich dataset, schools can identify patterns and correlations that might not be apparent through conventional means. This in-depth understanding is then combined with external information, such as current and projected social talent demands, industry development trends, and emerging job markets. By synthesizing internal student data with external market intelligence, institutions can provide students with highly personalized and relevant career planning advice.

Based on this foundation of comprehensive student profiling, schools can establish a robust and dynamic career planning support system. This system serves as a multifaceted platform, offering a suite of services designed to empower students in their career decision-making process. These services include advanced career assessment tools, real-time industry analysis, personalized employment information, and interactive career exploration modules. Through this comprehensive system, students can actively engage in their career planning process, querying and exploring suitable career directions that align closely with their individual characteristics and strengths.

The system provides detailed information about relevant job roles, including day-to-day responsibilities, required qualifications, and the work environment. Additionally, students can access up-to-date data on salary ranges and long-term career progression paths within their chosen fields. This enables students to formulate practical and achievable career development plans that are not static documents but dynamic roadmaps that can be adjusted as the student progresses through their education and as the job market evolves.

4 Evaluation of the Application Effects of Student Profiling Technology in Vocational Colleges

To comprehensively evaluate the application effects of student profiling technology in vocational colleges, we selected a vocational college as an empirical case study, analyzing the relevant data from 2019 to 2023. As shown in Figure 2.

The above charts show that since the application of student profiling technology began in 2019, the vocational college has achieved significant results in multiple areas:

The average student grades have shown a continuous upward trend, especially for the originally underperforming student groups.The school's academic early warning accuracy has improved by about 20 percentage points, allowing them to identify and help students with difficulties in a timely manner.Both the teaching and employment satisfaction of students have significantly improved, enhancing their learning enthusiasm and sense of recognition for the school. The student graduation rate has also improved, reflecting an overall improvement in their learning status.

5 Conclusions

The case study of the vocational college shows that the application of student profiling technology has achieved significant results in

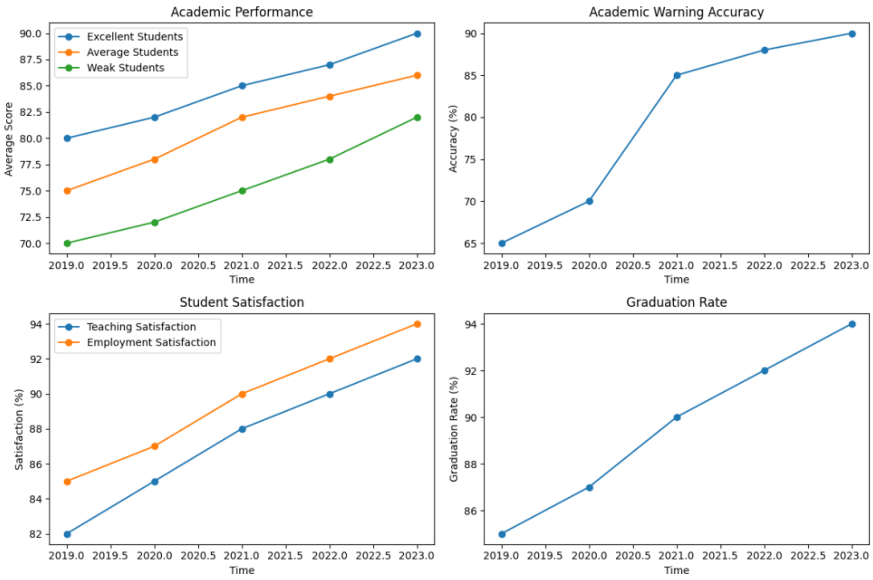


Figure 2: Evaluation of the Application Effects of Student Profiling Technology.

various aspects of talent cultivation. Based on data analysis and model application, students' academic performance, early warning accuracy, student satisfaction, and graduation rates have all been significantly improved.

Especially for originally weaker students, the improvement in academic performance indicates that the personalized teaching model based on student profiling can effectively meet the diverse learning needs of vocational college students. Improving the accuracy of the academic early warning system allows the school to provide timely and targeted support for students in difficulty, increasing their chances of success.From both the teaching and employment perspectives, the improvement in student satisfaction reflects the positive impact of student profiling technology in improving educational quality and career guidance services. The increase in graduation rates further demonstrates the overall improvement in the talent cultivation process of the vocational college.

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